

The ribbon operator is not a differential operator

$\text{ord}(N_e) = \infty$, and the order filtration is discontinuous at $t = -1$

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Abstract

Let $R_e(t)$ be the operator on symmetric functions that adds a connected border strip of size e with weight t^{ht} , let $g_e(t) = M[P_{(e)}(X; -t)]$ be multiplication by the one-row Hall–Littlewood polynomial, and let $E_e = g_e - R_e$ be the disconnected complement. Write $N_e := \partial_t E_e|_{t=-1}$, the operator that adds a size- e broken border strip with *exactly two* components, signed by height. We compute two closed forms for the iterated commutator of $R_e(t)$ with multiplication by p_1 , evaluated on two hook families, and deduce:

$$\text{ord}(R_e(c)) = \infty \text{ for every scalar } c \neq -1, \quad \text{ord}(R_e(-1)) = \text{ord}(M_{p_e}) = 0,$$

and $\text{ord}(N_e) = \text{ord}(E_e) = \infty$ for every $e \geq 2$. So N_e is not a derivation, is not of any finite order, and is not a differential operator on Λ at all. Consequently none of $R_e(t)$, $E_e(t)$, N_e lies in the algebra generated by multiplication operators M_f and skewing operators $f^\perp$. The mechanism is a single exceptional chain — the one that adds the strip to the empty partition — and the factor $(1+t)$ in the first closed form is exactly the two ways that chain can sit.

1 Setting and conventions

Fix the ground field $K = \mathbb{Q}(t)$ and let $\Lambda = K[p_1, p_2, \dots]$ be the ring of symmetric functions, with Schur basis $\{s_\lambda\}$. For $f \in \Lambda$ write M_f for multiplication by f and $f^\perp$ for its Hall-inner-product adjoint.

Definition 1 (order filtration). $\mathcal{D}^{\leq -1} = 0$, and for $d \geq 0$,

$$\mathcal{D}^{\leq d} = \{X \in \text{End}_K(\Lambda) : [X, M_f] \in \mathcal{D}^{\leq d-1} \text{ for all } f \in \Lambda\}.$$

Put $\text{ord}(X) = \min\{d : X \in \mathcal{D}^{\leq d}\}$, and $\text{ord}(X) = \infty$ if no such d exists.

This is Grothendieck’s filtration by differential order for the commutative ring Λ ; under $\Lambda = K[p_1, p_2, \dots]$ it is the order of X as a differential operator in the power sums. We use three standard facts.

Lemma 2. (i) $\text{ord}(M_f) = 0$ for every $f \in \Lambda$.

(ii) $\mathcal{D}^{\leq a} \mathcal{D}^{\leq b} \subseteq \mathcal{D}^{\leq a+b}$; in particular $\text{ord}(XY) \leq \text{ord}(X) + \text{ord}(Y)$, and $\mathcal{D}^{\leq d}$ is a two-sided $\mathcal{D}^{\leq 0}$ -module.

(iii) $\text{ord}(p_\rho^\perp) = \ell(\rho)$, and more generally $\text{ord}(f^\perp) = \deg_p f := \max\{\ell(\rho) : [p_\rho]f \neq 0\}$ (Macdonald I.5).

Lemma 3 (generators suffice). $X \in \mathcal{D}^{\leq d}$ if and only if $[\cdots [[X, M_{p_{a_1}}], M_{p_{a_2}}], \dots, M_{p_{a_{d+1}}}] = 0$ for all $a_1, \dots, a_{d+1} \geq 1$.

Proof. Necessity is immediate. For sufficiency, induct on d . The case $d = 0$ says $[X, M_{p_a}] = 0$ for all a ; since $[X, M_{fg}] = [X, M_f]M_g + M_f[X, M_g]$ and $[X, M_c] = 0$ for $c \in K$, an induction on the number of p -factors of a monomial p_ρ gives $[X, M_{p_\rho}] = 0$, hence $[X, M_f] = 0$ for all f by linearity, i.e. $X \in \mathcal{D}^{\leq 0}$. For $d \geq 1$ the same product rule, together with Lemma 2(ii) (which makes $\mathcal{D}^{\leq d-1}$ stable under left and right multiplication by M_g), propagates the hypothesis from generators to all f ; the inductive hypothesis then places $[X, M_f]$ in $\mathcal{D}^{\leq d-1}$. $\square$

Throughout we write

$$\text{ad}(X) := [X, M_{p_1}], \quad \text{ad}^d(X) = \sum_{r=0}^d (-1)^r \binom{d}{r} M_{p_1}^r X M_{p_1}^{d-r},$$

the second equality because the M_{p_1} commute with each other. By Lemma 3, if $\text{ad}^d(X) \neq 0$ for every d then $\text{ord}(X) = \infty$; this is the only direction we use.

1.1 The operators

Let $\lambda \subseteq \mu$ with $|\mu/\lambda| = e$. Call μ/λ a *broken border strip* if it contains no 2×2 square; let $m(\mu/\lambda)$ be its number of edge-connected components and $\text{ht}(\mu/\lambda) = \sum_{\text{components } c} (\# \text{rows}(c) - 1)$. Define

$$\begin{aligned} R_e(t) s_\lambda &= \sum_{m(\mu/\lambda)=1} t^{\text{ht}(\mu/\lambda)} s_\mu, \\ g_e(t) s_\lambda &= \sum_{\mu/\lambda \text{ broken}} (1+t)^{m(\mu/\lambda)-1} t^{\text{ht}(\mu/\lambda)} s_\mu, \quad E_e := g_e - R_e. \end{aligned}$$

At $t = -1$ the Murnaghan–Nakayama rule gives $R_e(-1) = M_{p_e}$ and $E_e(-1) = 0$.

Proposition 4 (the dictionary, Konvalinka; Halverson–Ram). $g_e(t) = M[P_{(e)}(X; -t)]$, multiplication by the one-row Hall–Littlewood polynomial at parameter $q = -t$. Explicitly, with $[n]_q = 1 + q + \cdots + q^{n-1}$,

$$P_{(e)}(X; -t) = \sum_{\rho \vdash e} \frac{(1+t)^{\ell(\rho)-1}}{z_\rho} \left(\prod_i [\rho_i]_{-t} \right) p_\rho. \quad (1)$$

In particular $\text{ord}(g_e) = 0$.

Derivation of (1). $P_{(r)}(X; q) = q_r(X; q)/(1-q)$ where $\sum_{r \geq 0} q_r(X; q)u^r = \prod_i \frac{1-qx_iu}{1-x_iu} = \exp(\sum_{n \geq 1} (1-q^n)p_n u^n/n)$. Put $\varepsilon := 1-q$, so $1-q^n = \varepsilon [n]_q$ and $\sum_r q_r u^r = \exp(\varepsilon B)$ with $B = \sum_n [n]_q p_n u^n/n$. Hence $P_{(e)} = \varepsilon^{-1} [u^e] \sum_{j \geq 1} \varepsilon^j B^j/j! = \sum_{j \geq 1} \varepsilon^{j-1} [u^e] B^j/j!$, and collecting compositions into partitions gives (1) with $q = -t$, $\varepsilon = 1+t$. The identification $g_e = M[P_{(e)}(X; -t)]$ is the quantum Murnaghan–Nakayama rule over broken ribbons. $\text{ord}(g_e) = 0$ by Lemma 2(i). $\square$

Set $\varepsilon := 1+t$. Since $t^h = (-1)^h(1-\varepsilon)^h$,

$$R_e = \sum_{D \geq 0} (-\varepsilon)^D \Psi_{e,D}, \quad \Psi_{e,D} s_\lambda = \sum_{m(\mu/\lambda)=1} \binom{\text{ht}(\mu/\lambda)}{D} (-1)^{\text{ht}(\mu/\lambda)} s_\mu, \quad (2)$$

so $\Psi_{e,0} = M_{p_e}$ and $\Psi_{e,1} = -\partial_t R_e|_{t=-1}$. Likewise $E_e = \varepsilon N_e + O(\varepsilon^2)$ with

$$N_e s_\lambda := \partial_t E_e|_{t=-1} s_\lambda = \sum_{m(\mu/\lambda)=2} (-1)^{\text{ht}(\mu/\lambda)} s_\mu, \quad (3)$$

because $(1+t)^{m-1}$ vanishes to order $m-1$ at $t=-1$: only $m=2$ survives the first derivative.

Remark 5. The definition (3) is *exactly two* components, not “at least two”. This corrects the statement of N_e in the session brief; the open-questions file has it right.

1.2 The merging expansion

Comparing ε -Taylor coefficients in $g_e = R_e + E_e$ using Proposition 4 and (2) gives, for each $D \geq 0$, an identity whose left side mixes component number with height and whose right side is a multiplication operator. We record the two we use.

Proposition 6. *For every $e \geq 2$,*

$$N_e - \Psi_{e,1} = M \left[\partial_t P_{(e)}(X; -t) \Big|_{t=-1} \right] = M \left[\frac{1}{2} \sum_{a=1}^{e-1} p_a p_{e-a} - \frac{e-1}{2} p_e \right],$$

an operator of order 0. Hence $\text{ord}(N_e) = \text{ord}(\Psi_{e,1})$ whenever either is positive, and $\text{ad}^d(N_e) = \text{ad}^d(\Psi_{e,1})$ for all $d \geq 1$.

Proof. Since $\text{ord}(g_e) = 0$ and the M_{p_a} do not depend on t , every t -Taylor coefficient of g_e is again a multiplication operator; the ε -linear one is $M[\partial_t P_{(e)}|_{t=-1}]$, and by (2), (3) it equals $-\Psi_{e,1} \cdot (-1) + N_e$, i.e. $N_e - \Psi_{e,1}$ up to the sign already absorbed in (2). For the explicit form, expand (1): $[n]_{-t} = n - \varepsilon \binom{n}{2} + O(\varepsilon^2)$, so writing $\mathbf{m}_\rho = \prod_i m_i(\rho)!$ and $z_\rho = \mathbf{m}_\rho \prod_i \rho_i$, the ρ -term of $P_{(e)}$ is $\frac{\varepsilon^{\ell(\rho)-1}}{\mathbf{m}_\rho} (1 - \varepsilon \frac{e-\ell(\rho)}{2} + O(\varepsilon^2)) p_\rho$. The ε^0 part is p_e ; the ε^1 part collects $\ell(\rho) = 2$ at leading order and $\rho = (e)$ at first order, giving $\frac{1}{2} \sum_{a=1}^{e-1} p_a p_{e-a} - \frac{e-1}{2} p_e$. $\square$

Remark 7 (why component number is only a filtration). Proposition 6 is the precise form of the statement that composing strip-additions merges components. The ε -linear part of g_e is *not* the two-component operator; it is the two-component operator minus the height-weighted connected operator $\Psi_{e,1}$. The discrepancy $\Psi_{e,1}$ is the merging term, and Theorem 10 below shows it is the term that carries all the operator content.

2 Two hook lemmas

Both closed forms are computed on hooks, where the poset of subshapes is a chain-like ladder and the enumeration is completely rigid. We use throughout: the coefficient of s_μ in $M_{p_1}^r X M_{p_1}^{d-r} s_\emptyset$ is

$$F_X(r) = \sum_{\nu \subseteq \kappa \subseteq \mu} f^\nu \langle s_\kappa, X s_\nu \rangle f^{\mu/\kappa}, \quad |\nu| = d-r, \quad |\mu/\kappa| = r, \quad (4)$$

where f^ν (resp. $f^{\mu/\kappa}$) is the number of standard Young tableaux of shape ν (resp. skew shape μ/κ), because M_{p_1} adds one box with coefficient 1. Then

$$\langle s_\mu, \text{ad}^d(X) s_\emptyset \rangle = \sum_{r=0}^d (-1)^r \binom{d}{r} F_X(r). \quad (5)$$

The right side annihilates every polynomial in r of degree $< d$. Each of the two lemmas below shows that $F_{R_e(t)}$ is such a polynomial *plus a spike at $r = d$* , and the spike is what survives.

Recall $f^{(a,1^b)} = \binom{a+b-1}{b}$, and that if a skew shape is a disjoint union of two connected chains (single rows or single columns) of sizes p and q then $f = \binom{p+q}{p}$.

Lemma 8 (arm-1 hooks). *Let $\mu = (2, 1^m)$. Its subshapes are exactly (1^j) , $0 \leq j \leq m+1$, and $(2, 1^j)$, $0 \leq j \leq m$. For $e \geq 2$, the pairs $\nu \subseteq \kappa \subseteq \mu$ with κ/ν a connected border strip of size e are exactly:*

- (A) $\nu = (1^{j'})$, $\kappa = (1^{j'+e})$, with $\text{ht} = e - 1$;
- (B) $\nu = \emptyset$, $\kappa = (2, 1^{e-2})$, with $\text{ht} = e - 2$;
- (C) $\nu = (2, 1^{j'})$, $\kappa = (2, 1^{j'+e})$, with $\text{ht} = e - 1$;

and those with κ/ν a broken border strip of size e with exactly two components are exactly

- (D) $\nu = (1^{j'})$ with $j' \geq 1$, $\kappa = (2, 1^{j'+e-2})$, with $\text{ht} = e - 2$.

Proof. A subshape of μ has first part ≤ 2 and all later parts ≤ 1 , which gives the list. If $\kappa = (1^j)$, $\nu = (1^{j'})$ the skew shape is the column-1 cells in rows $j' + 1, \dots, j$: one component, $j - j'$ rows. If $\kappa = (2, 1^j)$, $\nu = (2, 1^{j'})$ it is the column-1 cells in rows $j' + 2, \dots, j + 1$: again one component. If $\kappa = (2, 1^j)$ and $\nu = (1^{j'})$ the skew shape is the single cell $(1, 2)$ together with the column-1 cells in rows $j' + 1, \dots, j + 1$. For $j' = 0$ this set contains $(1, 1)$, which is edge-adjacent to $(1, 2)$: one component, $j + 1$ rows, and size $j + 2$; size e forces $j = e - 2$ and $\text{ht} = e - 2$, which is (B). For $j' \geq 1$ the column-1 cells all lie in rows ≥ 2 and are not edge-adjacent to $(1, 2)$: two components, of sizes 1 and $j + 1 - j'$, with $\text{ht} = 0 + (j - j')$; size e forces $j = j' + e - 2$ and $\text{ht} = e - 2$, which is (D). No shape here contains a 2×2 square, since column 2 carries at most one cell. Sizes and heights in (A),(C) are as stated. $\square$

Lemma 9 (hooks with arm e). *Let $\mu = (e, 1^d)$ with $e \geq 2$. The pairs $\nu \subseteq \kappa \subseteq \mu$ with κ/ν a connected border strip of size e are exactly:*

- (i) $\nu = \emptyset$, $\kappa = (e - b, 1^b)$ for $0 \leq b \leq \min(d, e - 1)$, with $\text{ht} = b$;
- (ii) $\nu = (a, 1^{b'})$ with $a \geq 1$, $\kappa = (a, 1^{b'+e})$, with $\text{ht} = e - 1$.

Proof. Subshapes of μ are $\emptyset$ and $(a, 1^b)$ with $1 \leq a \leq e$, $0 \leq b \leq d$. For $\nu = (a', 1^{b'})$ and $\kappa = (a, 1^b)$ the skew shape is the row-1 cells in columns $a' + 1, \dots, a$ together with the column-1 cells in rows $b' + 2, \dots, b + 1$. If $a' \geq 1$ both blocks avoid the corner and are mutually non-adjacent, so connectivity forces one of them to be empty; the row-1 block alone has size $a - a' \leq e - 1 < e$, so it is the column-1 block that survives, giving (ii) with $b - b' = e$. If $\nu = \emptyset$ the skew shape is all of $\kappa = (a, 1^b)$, a hook, which is connected, contains no 2×2 , has $b + 1$ rows and size $a + b$; size e gives (i) with $a = e - b \geq 1$. $\square$

3 The two closed forms

Theorem 10. *For all $e \geq 2$ and $d \geq 1$, with $\mu = (2, 1^{d+e-2})$,*

$$\langle s_\mu, \text{ad}^d(R_e(t)) s_\emptyset \rangle = (-1)^d t^{e-2} (1 + t).$$

Proof. Write $m = d + e - 2$, so $|\mu| = m + 2 = d + e$. We evaluate (4) for $X = R_e(t)$ using Lemma 8(A)–(C). In each case $r = |\mu| - |\kappa|$ and $|\nu| = d - r$.

(A) $\nu = (1^{j'})$, $\kappa = (1^{j'+e})$, so $r = (m + 2) - (j' + e) = d - j'$, i.e. $j' = d - r$. The constraint $\kappa \subseteq \mu$ reads $j' + e \leq m + 1 = d + e - 1$, i.e. $j' \leq d - 1$, i.e. $r \geq 1$. Here $f^\nu = 1$; and μ/κ consists of the single cell $(1, 2)$ together with $m + 1 - (j' + e)$ cells in column 1 starting at row $j' + e + 1 \geq 3$, two disjoint chains, so $f^{\mu/\kappa} = \binom{1+(m+1-j'-e)}{1} = m + 2 - j' - e = d - j' = r$. The weight is t^{e-1} . Total: $F_A(r) = r t^{e-1}$ for $1 \leq r \leq d$, and $F_A(0) = 0$, so $F_A(r) = r t^{e-1}$ for all $0 \leq r \leq d$.

(B) $\nu = \emptyset$, $\kappa = (2, 1^{e-2})$, $|\kappa| = e$, so $r = d$ and $|\nu| = 0$, consistent. Here $f^\nu = 1$, μ/κ is the column-1 chain in rows $e, \dots, m + 1$ so $f^{\mu/\kappa} = 1$, and the weight is t^{e-2} . Thus $F_B(r) = t^{e-2} \llbracket r = d \rrbracket$.

(C) $\nu = (2, 1^{j'})$, $\kappa = (2, 1^{j'+e})$, so $r = (m + 2) - (j' + e + 2) = d - 2 - j'$. The constraint $\kappa \subseteq \mu$ reads $j' + e \leq m$, i.e. $j' \leq d - 2$, i.e. $r \geq 0$; and $j' \geq 0$ gives $r \leq d - 2$. Here $f^\nu = f^{(2, 1^{j'})} = j' + 1 = d - 1 - r$, μ/κ is a single column chain so $f^{\mu/\kappa} = 1$, and the weight is t^{e-1} . Thus $F_C(r) = (d - 1 - r) t^{e-1}$ for $0 \leq r \leq d - 2$ and 0 otherwise.

Adding, for $r \leq d - 2$ we get $(r + (d - 1 - r)) t^{e-1} = (d - 1) t^{e-1}$; for $r = d - 1$ we get $(d - 1) t^{e-1}$; and for $r = d$ we get $d t^{e-1} + t^{e-2} = (d - 1) t^{e-1} + t^{e-2}(1 + t)$. Hence

$$F_{R_e}(r) = (d - 1) t^{e-1} + t^{e-2}(1 + t) \llbracket r = d \rrbracket.$$

Now apply (5): the constant term is annihilated because $\sum_r (-1)^r \binom{d}{r} = 0$ for $d \geq 1$, and the spike contributes $(-1)^d \binom{d}{d} t^{e-2}(1 + t)$. $\square$

Theorem 11. For all $e \geq 2$ and $d \geq 1$, with $\mu = (e, 1^d)$,

$$\langle s_\mu, \text{ad}^d(R_e(t)) s_\emptyset \rangle = (-1)^d \left[\sum_{b=0}^{\min(d, e-1)} \binom{d}{b} t^b - \binom{d-1}{e-1} t^{e-1} \right].$$

Proof. Here $|\mu| = e + d$. Use Lemma 9.

(i) $\nu = \emptyset$, $\kappa = (e - b, 1^b)$, $|\kappa| = e$, hence $r = d$ and $|\nu| = 0$. Then $f^\nu = 1$ and μ/κ consists of the b row-1 cells in columns $e - b + 1, \dots, e$ and the $d - b$ column-1 cells in rows $b + 2, \dots, d + 1$, disjoint chains, so $f^{\mu/\kappa} = \binom{d}{b}$. The weight is t^b . Thus case (i) contributes $\llbracket r = d \rrbracket \sum_{b=0}^{\min(d, e-1)} \binom{d}{b} t^b$.

(ii) $\nu = (a, 1^{b'})$, $\kappa = (a, 1^{b'+e})$, so $r = (e + d) - (a + b' + e) = d - a - b'$; write $s := a + b' = d - r$. Then $f^\nu = \binom{s-1}{b'} = \binom{s-1}{a-1}$, and μ/κ consists of the $e - a$ row-1 cells in columns $a + 1, \dots, e$ and the $d - b' - e$ column-1 cells in rows $b' + e + 2, \dots, d + 1$, so $f^{\mu/\kappa} = \binom{r}{e-a}$ — an expression that vanishes automatically when $a > e$ or when $b' + e > d$, so no further constraints are needed. The weight is t^{e-1} . For $s \geq 1$, Vandermonde gives

$$\sum_{a \geq 1} \binom{s-1}{a-1} \binom{r}{e-a} = \sum_{i \geq 0} \binom{s-1}{i} \binom{r}{e-1-i} = \binom{s-1+r}{e-1} = \binom{d-1}{e-1},$$

a constant independent of r . For $s = 0$, i.e. $r = d$, the sum is empty and equals 0. Hence

$$F_{R_e}(r) = \binom{d-1}{e-1} t^{e-1} + \llbracket r = d \rrbracket \left[\sum_{b=0}^{\min(d, e-1)} \binom{d}{b} t^b - \binom{d-1}{e-1} t^{e-1} \right],$$

and (5) kills the constant and keeps $(-1)^d$ times the bracket. $\square$

Remark 12. The two theorems agree where the two hook families coincide. At $e = 2$ both μ 's are $(2, 1^d)$, and Theorem 11 gives $(-1)^d[1 + dt - (d-1)t] = (-1)^d(1+t)$, which is Theorem 10 at $e = 2$.

Remark 13 (the mechanism). In both proofs $F_{R_e}(r)$ is a polynomial in r of degree $< d$ plus a spike at $r = d$, and the spike is the contribution of the chains with $\nu = \emptyset$: those in which the whole e -ribbon is added to the *empty* partition. This is exactly the locus where “connected” and “broken” disagree, because the corner cell $(1, 1)$ glues the arm to the leg. The alternating sum $\sum_r (-1)^r \binom{d}{r}$ is blind to polynomials and sees only the spike, so *the whole theorem is one exceptional chain*. In Theorem 10 the spike is $t^{e-2} + t^{e-1}$: the ribbon added to $\emptyset$ can sit as the hook $(2, 1^{e-2})$ (height $e-2$) or, as the failure of case (C) at $r = d$, as the column (1^e) (height $e-1$). *The factor $(1+t)$ is those two seats.*

4 Consequences

Corollary 14. *Let c be a scalar. Then $\text{ord}(R_e(c)) = 0$ if $c = -1$, and $\text{ord}(R_e(c)) = \infty$ for every $c \neq -1$. In particular $\text{ord}_{\mathbb{Q}(t)}(R_e(t)) = \infty$ for every $e \geq 2$.*

Proof. $R_e(-1) = M_{p_e}$ has order 0. For $c \notin \{0, -1\}$, Theorem 10 gives $\text{ad}^d(R_e(c)) \neq 0$ for every $d \geq 1$, since $(-1)^d c^{e-2}(1+c) \neq 0$. For $c = 0$, Theorem 11 gives $(-1)^d \binom{d}{0} = (-1)^d \neq 0$ (all other terms carry a positive power of t , and $e-1 \geq 1$). Apply Lemma 3. $\square$

Corollary 15. *For every $e \geq 2$, $\text{ord}(\Psi_{e,1}) = \text{ord}(N_e) = \text{ord}(E_e) = \infty$. Explicitly, for all $d \geq 1$,*

$$\langle s_{(2, 1^{d+e-2})}, \text{ad}^d(N_e) s_{\emptyset} \rangle = (-1)^{d+e+1}.$$

In particular N_e is not a derivation, is not of finite order, and is not a differential operator on Λ .

Proof. The operators M_{p_1} do not depend on t , so ad commutes with ∂_t . Applying $-\partial_t|_{t=-1}$ to Theorem 10 and using $\Psi_{e,1} = -\partial_t R_e|_{t=-1}$ gives

$$\langle s_{(2, 1^{d+e-2})}, \text{ad}^d(\Psi_{e,1}) s_{\emptyset} \rangle = -(-1)^d \partial_t(t^{e-2}(1+t)) \Big|_{t=-1} = -(-1)^d (-1)^{e-2} = (-1)^{d+e+1},$$

where the middle evaluation holds because $(1+t)$ vanishes at $t = -1$, so only the term in which ∂_t hits the factor $(1+t)$ survives, leaving $t^{e-2}|_{t=-1}$. By Proposition 6, $\text{ad}^d(N_e) = \text{ad}^d(\Psi_{e,1})$ for $d \geq 1$, which gives the displayed value; it is never zero, so $\text{ord}(N_e) = \infty$ by Lemma 3. A derivation has order ≤ 1 , so N_e is not one. Finally $E_e = \varepsilon N_e + O(\varepsilon^2)$ and ord of every t -Taylor coefficient of E_e is bounded by $\text{ord}_{\mathbb{Q}(t)}(E_e)$, so $\text{ord}(E_e) = \infty$; alternatively $E_e = g_e - R_e$ with $\text{ord}(g_e) = 0$ and Corollary 14. $\square$

Corollary 16 (no Pieri presentation). *Let $\mathcal{A} \subseteq \text{End}_K(\Lambda)$ be the (associative, unital) algebra generated by all multiplication operators M_f and all skewing operators $f^\perp$, $f \in \Lambda$. Then for every $e \geq 2$ and every scalar $c \neq -1$,*

$$R_e(c) \notin \mathcal{A}, \quad E_e(c) \notin \mathcal{A}, \quad N_e \notin \mathcal{A}.$$

Proof. By Lemma 2, $\text{ord}(M_f) = 0$ and $\text{ord}(f^\perp) = \deg_p f < \infty$, and order is subadditive under composition and takes the max under sums; so every element of $\mathcal{A}$ has finite order. Corollaries 14 and 15 say these operators do not. $\square$

- Remark 17** (what this settles for the $(1+t)$ programme). 1. The route to the sharpness of $(1+t)^{k-1}$ through “is N_e a derivation?” is closed, and closed in the strongest direction: N_e is not merely a non-derivation, it has no finite order at all. The predicate $j_{\min} \geq \lceil (k-1)/d \rceil$ with $d = \text{ord } N_e$ degenerates to $j_{\min} \geq 1$, which is consistent with — and gives no new information beyond — the independently proved sharpness statement.
2. $\text{ord}(g_e) = 0$ while $\text{ord}(R_e) = \text{ord}(E_e) = \infty$: the decomposition of a multiplication operator into its connected and disconnected parts destroys finite order. This is a quantitative form of “a truncation is not a multiplication operator”, and it explains structurally why the commutator of two such truncations is not addressed anywhere in the Hall–Littlewood literature: the objects are not differential operators, so none of the standard machinery for M_f and $f^\perp$ applies to them.
3. Corollary 16 *separates* the ribbon operators from the algebra generated by Pieri operators and their transposes, rather than reducing one to the other. Any question posed inside that algebra — e.g. the commutation relations between $h_{i\cdot}$ and $h_j^\perp$ — lives entirely in finite order and therefore cannot have R_e or N_e among its consequences.
4. The order filtration of $R_e(t)$ is *discontinuous at the anchor*: 0 at $t = -1$, ∞ everywhere else. Every measurement made at $t = -1$ is therefore blind to the operator content of the family.

Remark 18 (the bilinearity identity, for the record). Since $g_e = R_e + E_e$ and $[g_e, g_{e'}] = 0$ (multiplication operators in a commutative ring, Proposition 4), bilinearity of the bracket gives

$$[R_e, R_{e'}] = -([R_e, E_{e'}] + [E_e, R_{e'}] + [E_e, E_{e'}]).$$

This is one line and it cannot fail; it carries no information beyond $[g_e, g_{e'}] = 0$, which is itself forced. Its only genuine consequence is the ε^0 statement: both sides vanish at $t = -1$ because $E_e(-1) = 0$ and $R_e(-1) = M_{p_e}$, which re-derives $(1+t) \mid [R_e, R_{e'}]$ in one step. It is recorded here so that it is not mistaken for content.

Remark 19 (a reduction whose two sides are both infinite). A previous session recorded, from $\text{ord}(g_e) = 0$, that $\text{ord}(R_e) = \text{ord}(E_e)$ and concluded that “the operator content lives entirely in the disconnected part”. The identity is correct and is reproved by Corollary 15. But both sides are now known to be ∞ , so the identity bounds nothing: it is a true equation between two infinities, not a reduction. What survives of that observation is the sharper Corollary 16.

5 Verification

All computations are in `probes/2026-09-06-Q84/`. Two engines were used and kept code-disjoint: `engine.py` enumerates skew shapes directly and works over $\mathbb{Q}(t)$ with `sympy`; `fast.py` works at $t = -1$ (or at integer t) with the Maya/abacus bead model for connected ribbons and a constructive row-increment enumeration for broken ones.

1. R_e *against prior code*. `engine.op_R` agrees with `probes/2026-09-04-Q76/abacus.py` on all 225 pairs (λ, e) with $|\lambda| \leq 7$, $1 \leq e \leq 5$.

2. *Proposition 4.* $M[P_{(e)}(X; -t)]s_\lambda = g_e s_\lambda$ on all 150 pairs with $1 \leq e \leq 5$, $|\lambda| \leq 6$, exactly, with no free scalar: the left side is computed from (1) by iterated Murnaghan–Nakayama, the right side from broken-strip enumeration.
3. *Proposition 6.* $N_e - \Psi_{e,1} = M[\cdots]$ on all 335 pairs with $2 \leq e \leq 6$, $|\lambda| \leq 8$.
4. $N_{e\emptyset} = 0$, checked (not assumed) for $e = 3, 4, 5$.
5. *Theorem 10.* 245/245 agreements over $e \in \{2, \dots, 6\}$, $d \in \{1, \dots, 7\}$, $t \in \{-3, -2, -1, 0, 1, 2, 4\}$; and the symbolic version over $\mathbb{Q}(t)$ for $e \leq 4$, $d \leq 3$.
6. *Corollary 15.* $(-1)^{d+e+1}$ confirmed on 40/40 pairs $e \in \{2, \dots, 6\}$, $d \in \{1, \dots, 8\}$, computed from the two-component definition (3) directly — a code path disjoint from the one used for Theorem 10.
7. *Theorem 11.* 280/280 agreements over $e \in \{2, \dots, 6\}$, $d \in \{1, \dots, 7\}$, $t \in \{-3, \dots, 5\}$.
8. *Negative controls* (`controls.py`). The same bracket harness, run on operators of known order, vanishes at exactly the right depth: M_{h_2} (order 0) has $C^{(1)} = 0$; $M_{p_1 p_1^\perp}$ (order 1) has $C^{(1)} \neq 0$, $C^{(2)} = 0$; $M_{p_3 p_1 p_1^\perp}$ (order 2, degree +3, structurally parallel to N_3) has $C^{(2)} \neq 0$, $C^{(3)} = 0$. The sharpest control is the minimal pair: M_{h_2} and N_2 are computed by the same two-box enumeration with one predicate flipped (horizontal strip versus two components), and have orders 0 and ∞ .
9. *Full sweep.* $C^{(d)} \neq 0$ for N_e , $e \in \{2, 3, 4\}$, $d \in \{1, 2, 3\}$, over all ordered a -tuples in $\{1, 2, 3\}^d$ (`itertools.product`, not `combinations`) and all λ with $|\lambda| \leq 6$: 3481 of 3510 triples $(e, \mathbf{a}, \lambda)$ give a nonzero result. The 29 vanishing triples are individual $(\lambda, \mathbf{a})$ coincidences spread over $|\lambda| \in \{1, 3, 5, 6\}$, not a pattern; what is at stake is the vanishing of an *operator*, and Theorems 10 and 11 exhibit states where it does not vanish, for every d .

6 Gaps and what is not claimed

1. Corollary 14 is stated for scalars c in a field of characteristic 0; the two hook families jointly cover $c \neq -1$, but a single uniform witness family covering all $c \neq -1$ at once is not given.
2. Nothing here computes $\text{ord}(\Psi_{e,D})$ for $D \geq 2$. The natural guess — 0 for $D = 0$ and ∞ for every $D \geq 1$ — is untested.
3. Proposition 4 is used only through $\text{ord}(g_e) = 0$ and the ε -linear coefficient of (1); both are verified independently above. The attribution of $g_e = M[P_{(e)}(X; -t)]$ to the quantum Murnaghan–Nakayama rule rests on a prior reading session, but the identity itself is derived here from the Hall–Littlewood generating function and checked, so no step of the present argument depends on the attribution.
4. Remark 13(3) refers to the algebra generated by Pieri operators and their transposes. What is proved is the mathematical statement Corollary 16; the reading of it as an answer to a particular posed question is a remark, not a theorem, and the source of that question was not re-read in this session.